

Project Goal

Goal: Validate and improve the existing Integrated Sensor Comfort System prototype and develop a testing protocol to find correlation between reported long-term comfort and quantitative parameters.

Parameters: Pressure force, frictional force along the thigh, vibration transmissibility

Background

Current solutions for long-term automotive seat comfort testing: human testing (contains bias) or manikin testing (non-representative) (Figure 1).

Improvements to be made to the existing prototype (Figure 2):

- More human-like tissue mimicking layer
- Increased durability of circuitry
- Addition of sensors for vibration transmissibility
- Revise sensor layout to improve accuracy

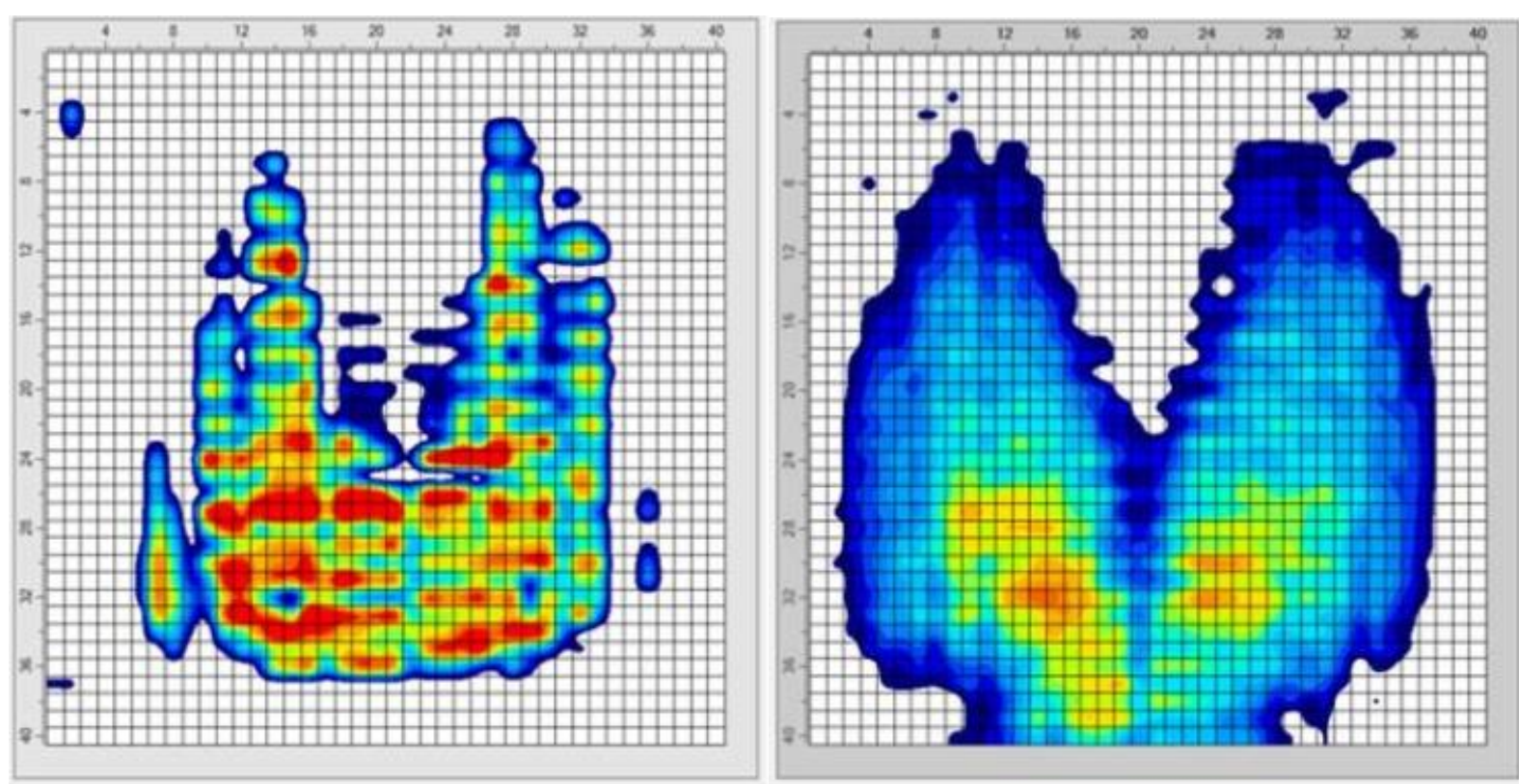


Figure 1: Manikin (left) seat pressure map versus human (right) seat pressure map

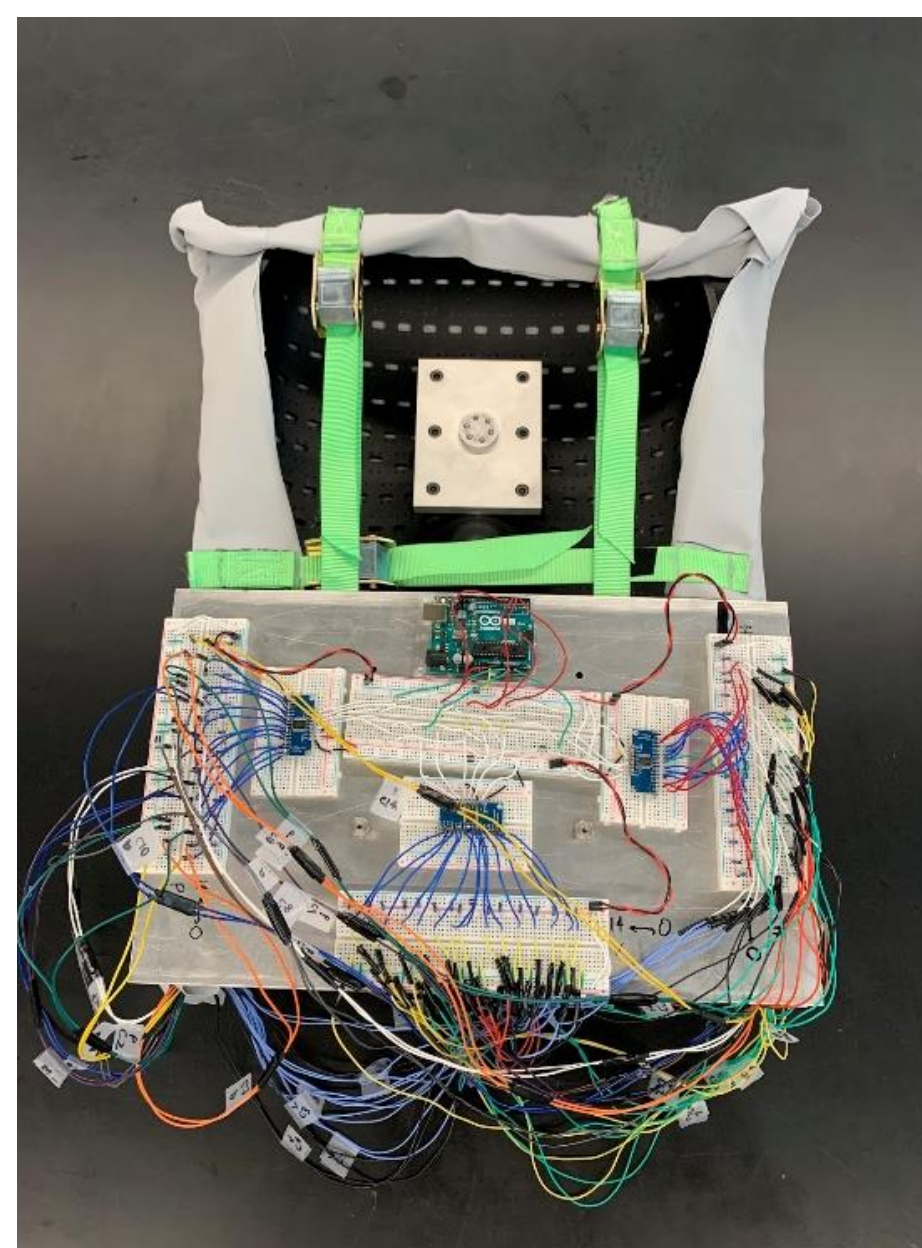


Figure 2: Integrated Sensor Comfort System 1.0

Device Design

Tissue Mimicking Layer:

- Consists of a layer of Smooth-On Silicone
- Varying thickness from thigh and gluteal regions to represent different pressure due to bone and tissue
- Load-deflection testing performed on samples of varying thickness and compared to human data to determine thickness.

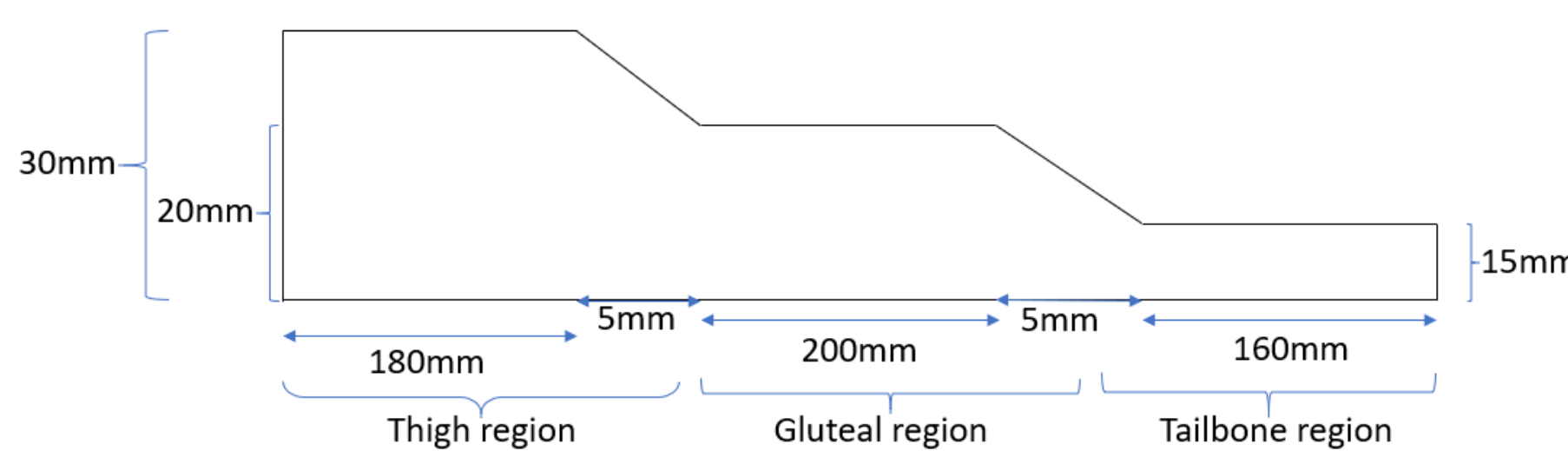


Figure 3: New tissue mimicking layer design



Figure 4: Comparison of the previous year's tissue layer (A) and the new variable thickness tissue layer (B)

Sensor Layers:

- Pressure data measured by fifteen ohmite pressure sensors
- Frictional force data measured by 30 rubber cord stretch sensors
- New "L" configuration
- Non-conductive fabric between sensor layers
- Accelerometers placed on the base of the seat and on the circuitry platform of the prototype

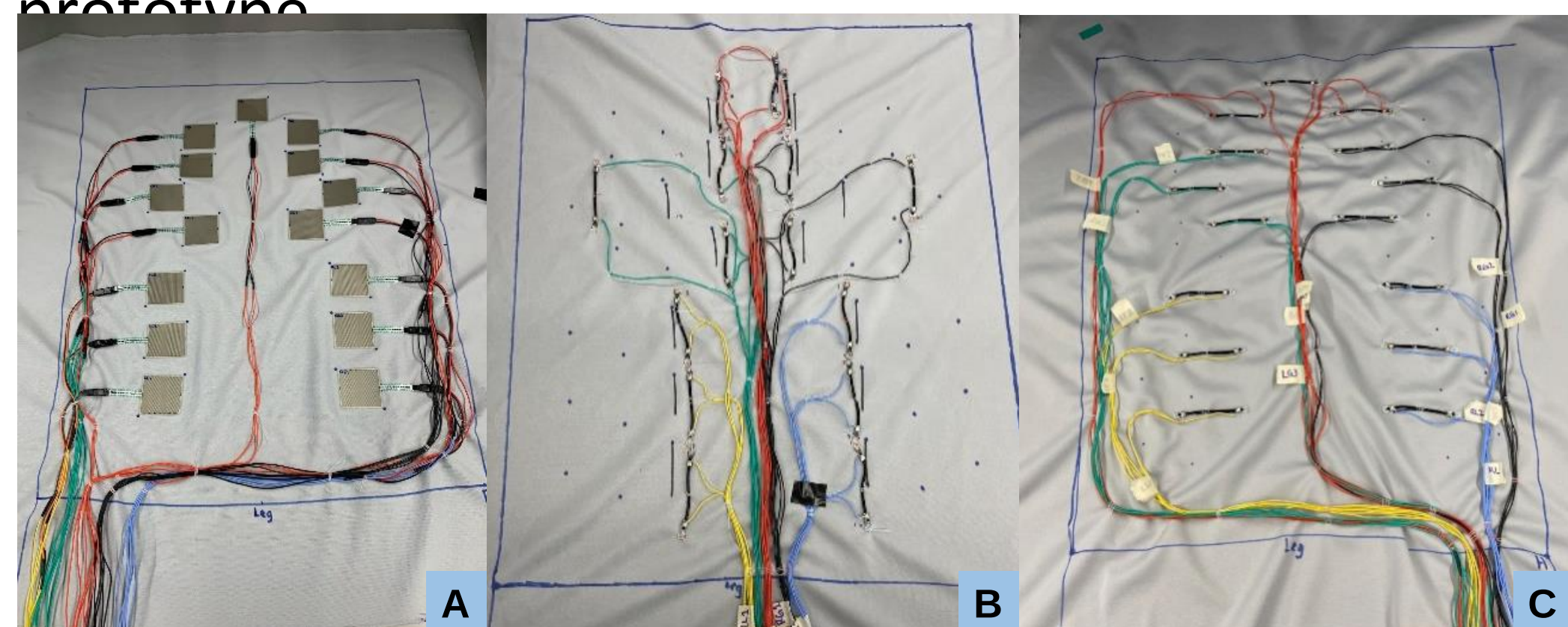


Figure 5: Pressure (A), vertical (B), and horizontal (C), sensor layers.

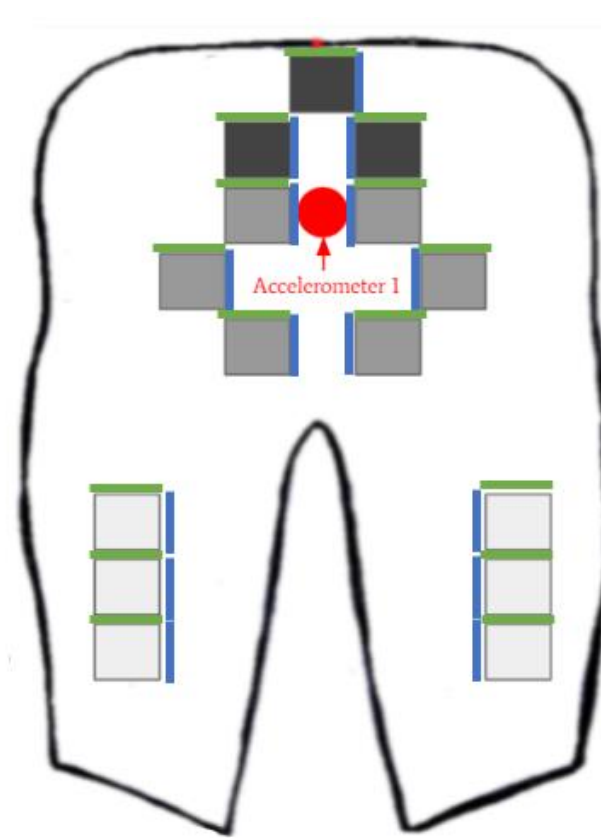


Figure 6: Node layout

Circuitry:

- An Arduino Uno Board is connected to the circuitry
- 3 multiplexers
- 45 voltage divider circuits
- All wires were soldered to protoboards for durability of the device

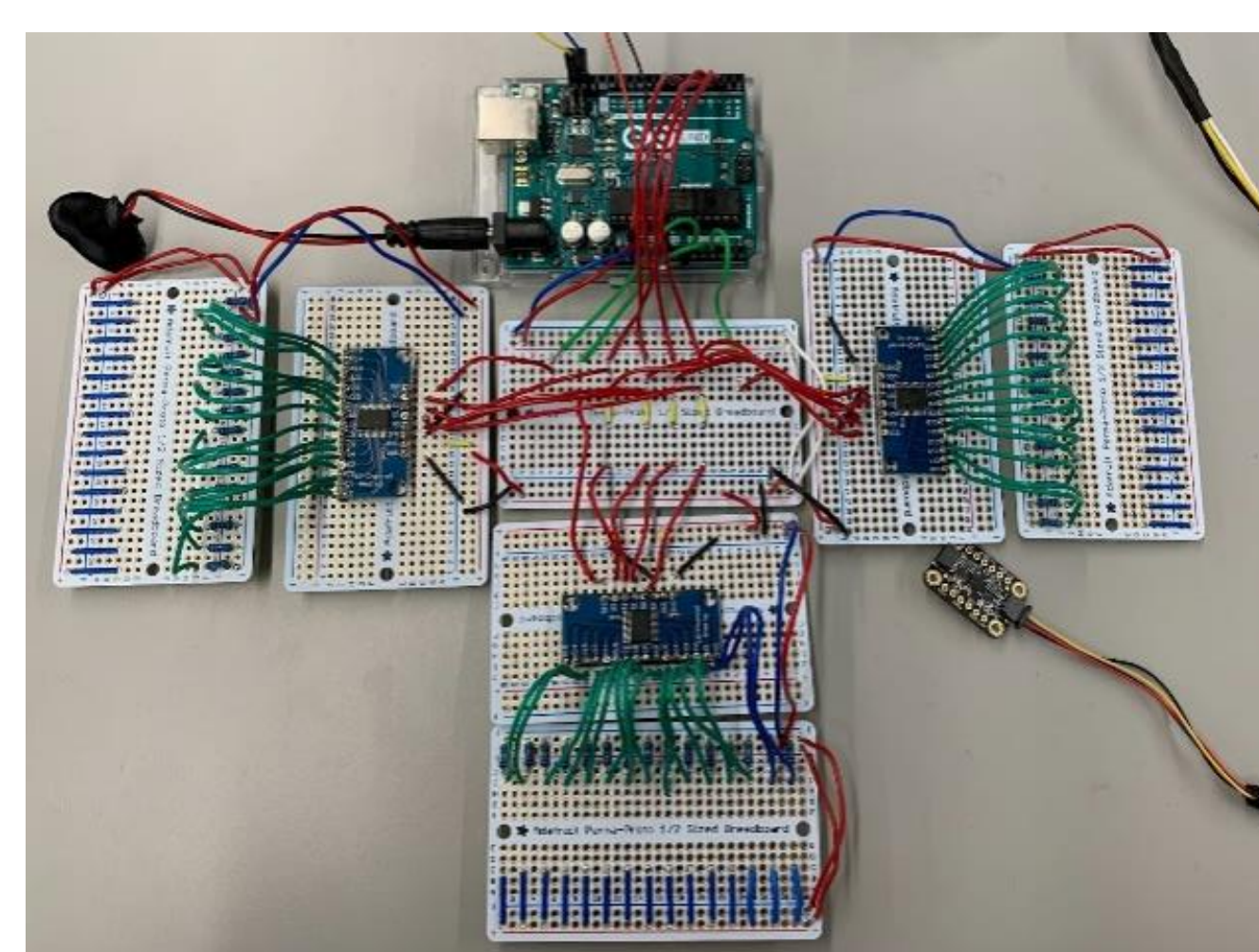


Figure 7: Circuitry platform that includes 7 protoboards, 3 multiplexers 45 voltage divider circuits

Coding:

- Used Arduino IDE to run the code that scans 15 nodes from 3 multiplexers, and reads accelerometer readings, then processes the data and outputs pressure

Testing and Results

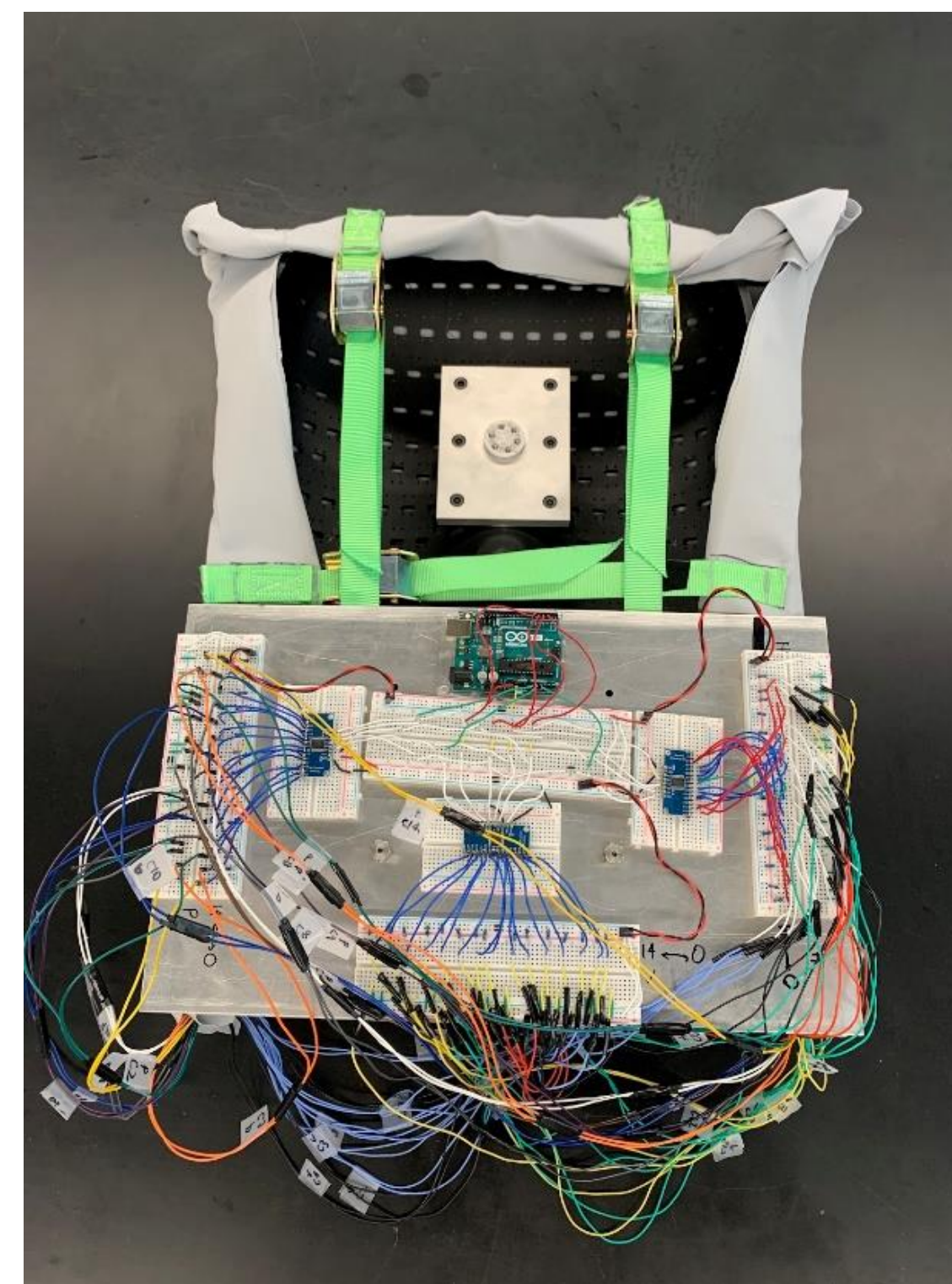


Figure 8: Full scale Integrated Sensor Comfort System 2.0 prototype
****Replace w/ new prototype****

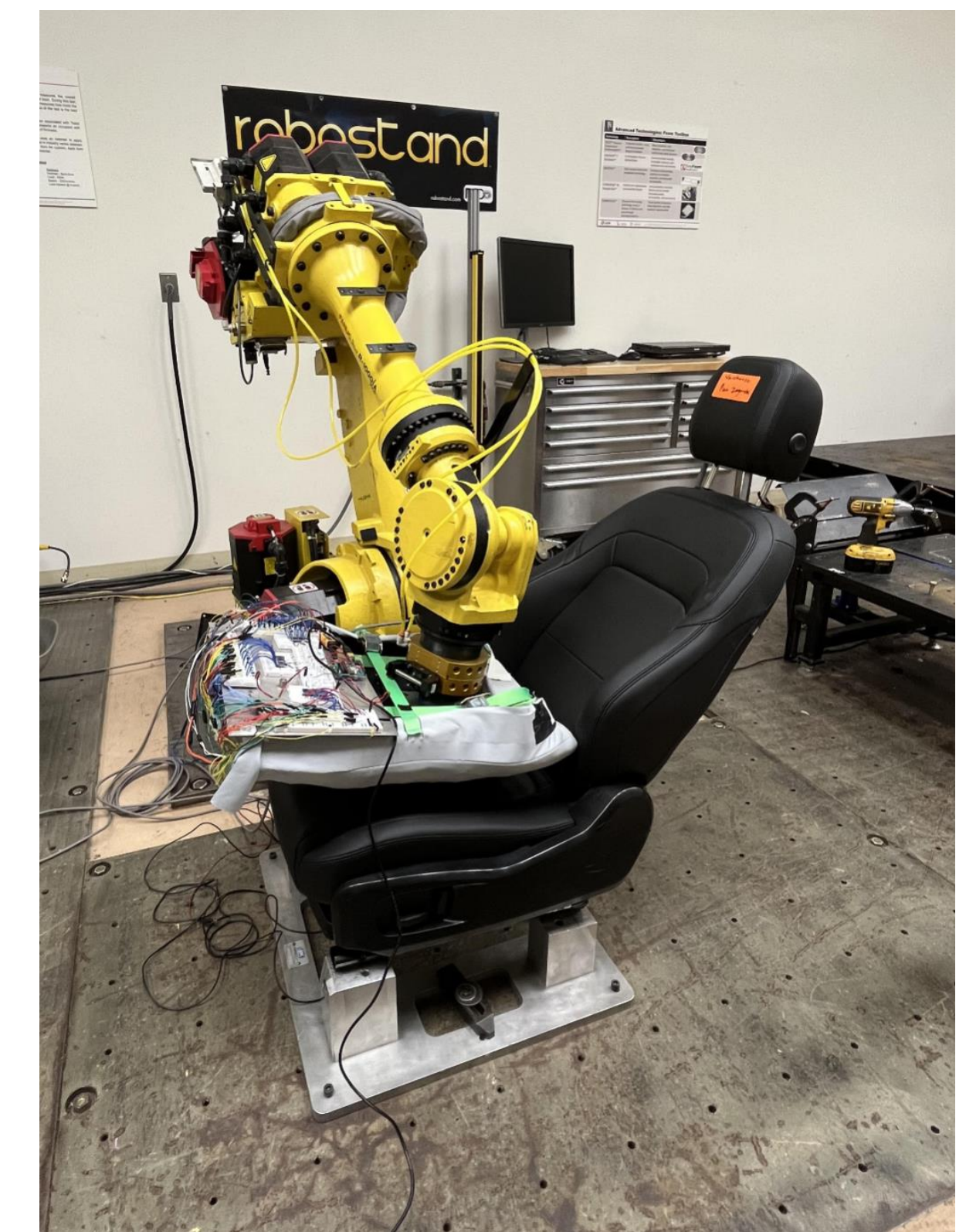


Figure 9: Compression testing using the KUKA robot

Compression testing: Robot arm presses down on a seat and sensor readings are recorded at an interval. The results are shown as pressure maps, comparing last year and this year's results for accuracy (Figure 10)



Figure 10: Pressure maps showing accuracy of new device to existing device

Vibration transmissibility: Acceleration ratio of seat to base at discrete frequencies

- MATLAB code processed the raw data from 2 Adafruit LIS331 triaxial accelerometers and calculated vibrational transmissibility (Figure 11)
- Accuracy of the accelerometers and coding was tested in the Lear ComforTec Lab (Figure 12)
- Peak transmissibility values from both the shaker table and MATLAB code were found to be similar at 3.5 Hz and 3.68 Hz, respectively

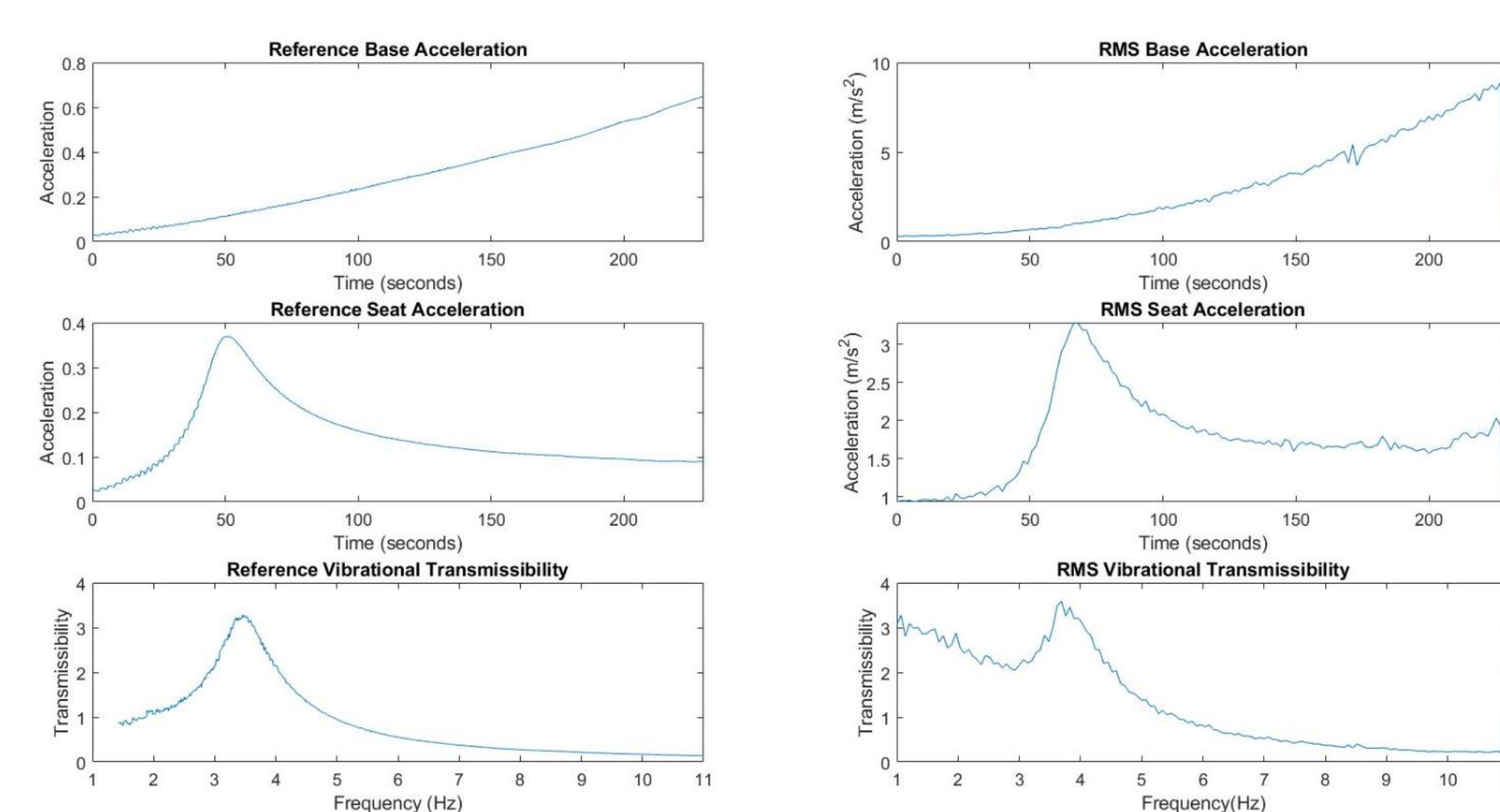


Figure 11: Comparison of shaker table reference data (left column) and the experimental data (right column)

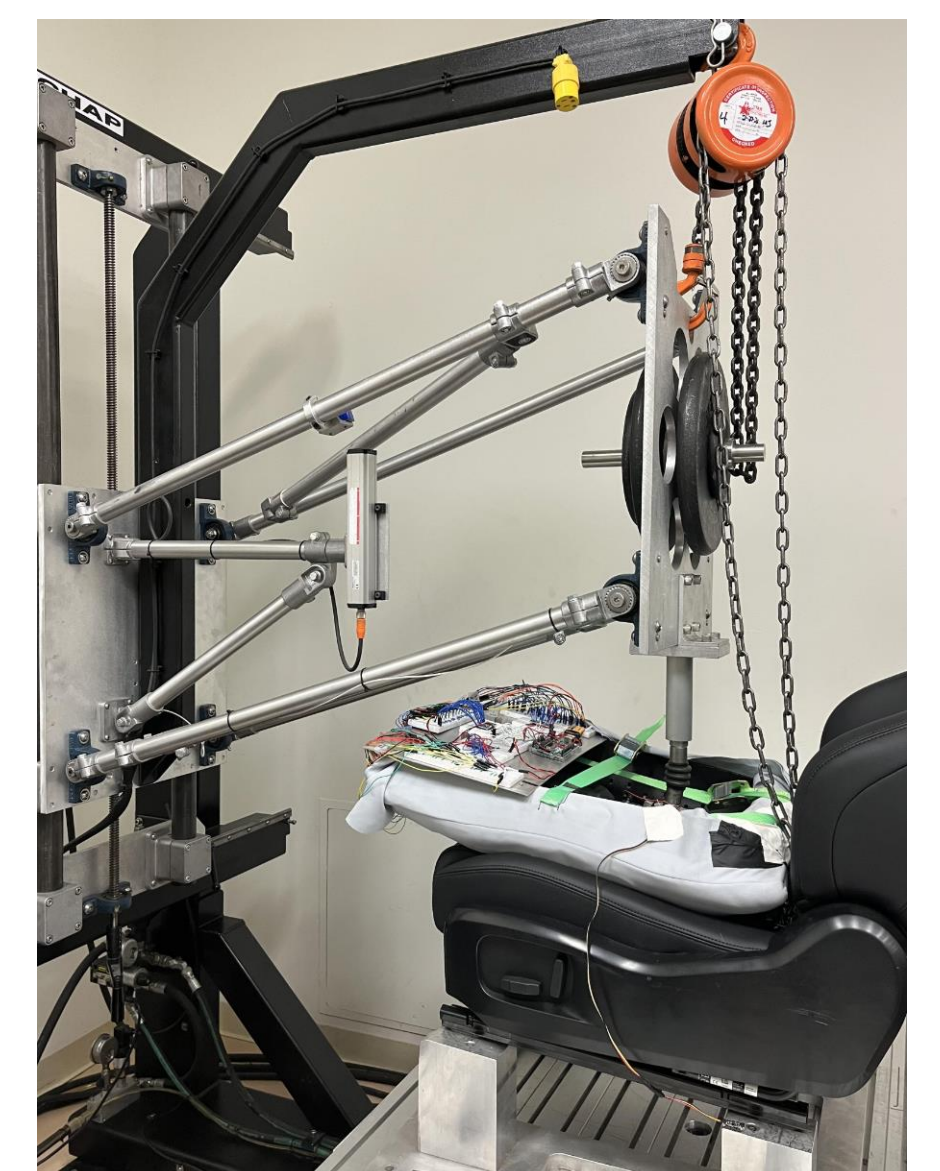


Figure 12: Shaker table testing to assess accuracy of vibration transmissibility sensors and code

Conclusion and Future Work

- Variable tissue layer yielded more accurate readings
- New orientation of sensors improved accuracy of readings
- Accelerometers and MATLAB code accurately calculated transmissibility
- Soldering wires and adding an enclosure improved durability

Improvement of Tissue Layer:

- Develop various tissue layers to cater to different percentile males and females

Improvement of Sensors:

- Streamline the sensor calibration process
- Search for stretch sensors with faster recovery time

Improvement of Circuit:

- PCB to decrease amount of wires
- Microcontroller with bluetooth or memory

Acknowledgements

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